

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the definite integral $\int_1^3 3x \, dx$.
 - a) Sketch what is graphically represented by this integral.
 - b) Use geometry to calculate this integral.
 - c) Show that this integral equals 12 using the limit definition of the definite integral.
 - d) Show that this integral equals 12 using the limit definition of the definite integral and where you use left endpoints for your selection of x_i^* .
 - e) TRUE or FALSE: If the function $y = 3x$ is integrable on $[1, 3]$ then it must always produce a integral value of 12 regardless of the selection of the points for x_i^* .
 - f) Based on your answer to the previous part, how could one ever possibly show that a function is *NOT* integrable?

2. Evaluate the integral $\int_0^1 (x^3 - 3x^2) \, dx$ using the limit definition of the definite integral.
3. Relate each of the following limits to a definite integral over the specified interval. (Use right endpoints.)

a) $\lim_{n \rightarrow \infty} \frac{\sqrt{1} + \sqrt{2} + \sqrt{3} + \dots + \sqrt{n}}{n\sqrt{n}}$ over $[0, 1]$.

b) $\lim_{n \rightarrow \infty} \pi \left(\frac{\sin\left(\frac{2\pi}{n}\right) + \sin\left(\frac{4\pi}{n}\right) + \sin\left(\frac{6\pi}{n}\right) + \dots + \sin\left(\frac{2n\pi}{n}\right)}{n} \right)$ over $[0, \pi]$.

c) $\lim_{n \rightarrow \infty} \frac{1^7 + 2^7 + 3^7 + \dots + n^7}{n^8}$ over $[0, 1]$.

4. Consider the integral $\int_1^2 x^{-2} \, dx$.

- a) Construct a limit (where x_i^* represents the right endpoints of our sub-intervals) that is equal to this integral.
- b) In the previous questions we were always using x_i^* as the right endpoint of our intervals. This worked well for these examples, but sometimes it can be easier to use other values. Show how if we choose x_i^* to be the geometric mean of x_{i-1} and x_i (that is, $x_i^* = \sqrt{x_{i-1} \cdot x_i}$) then we can find $\int_1^2 x^{-2} \, dx$.

Hint: determine some formulas for x_{i-1} & x_i and use the identity $\frac{1}{m(m+1)} = \frac{1}{m} - \frac{1}{m+1}$.